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Old Chinese: A New Reconstruction

Nathan W. Hill

Baxter, H. William, and Laurent Sagart. *Old Chinese: A New Reconstruction*. Oxford University Press, 2014. xvii + 431 pp. ISBN 9780199945375. Price 78.00 USD.

New ideas about the reconstruction of Old Chinese catch on slowly. Although in 1954 André-Georges Haudricourt explained the origin of Middle Chinese tones as arising from lost segments,¹ Li Fang-Kuei's failure to endorse Haudricourt's explanation² delayed acceptance of this proposal; even in 2014, some investigators remain unaware of Haudricourt's work.³ Sergei Jaxontov, an early adherent of Haudricourt's explanation, combined the latter's proposal with a hypothesis of Arisaka Hideyo⁴ and two of his own ideas⁵ to yield a seven vowel version of Old Chinese, but with vowels of idiosyncratic distribution.⁶ In a lecture delivered at Princeton University in 1971, Nicholas Bodman modified the system of Jaxontov to yield six vowels with more balanced distribution; Bodman's student William

¹ A. G. Haudricourt, "Comment reconstruire le chinois archaïque," 351–64. Conrady, "Alte westöstliche Kulturwörter," 14, earlier conjectured that the departing tone (去聲) derives from *-s.

² Li Fang-Kuei, 上古音研究, 24–26.

³ Song, "Stop codas in Old Chinese and Proto Sino-Tibetan," 96–135.

⁴ The "front vowel hypothesis" holds that division four (四等) words originate from front vowels rather than a palatal medial. See Arisaka, カルグレン氏の拗音節を評す, 354–55, and Baxter, *Handbook*, 240–47.

⁵ The "r-hypothesis," proposed by Jaxontov, "Consonant combinations," 2–9, but presaged by von der Gabelentz, *Chinesische Grammatik*, 99, §223, accounts for the origins of second division (二等) words with a medial -r-. Jaxontov originally proposed medial -l-, but subsequent researchers have generally amended this to -r- (see Baxter, *Handbook*, 262). E. Pulleyblank, "The Consonantal System of Old Chinese," 111–14, accepted this proposal, and added to it the "rj-hypothesis," that *chóngniǔ* division three (重紐三等) words also originally had a medial -r- (see Baxter, *Handbook*, 280–88). Jaxontov, "Фонетика," 104 (see Baxter, *Handbook*, 236–40), also first articulated the "rounded vowel hypothesis," that Middle Chinese -w- results from the breaking of rounded vowels before dentals, or the re-phonemization of labiovelar initials.

⁶ Jaxontov, *Древнекитайский язык*, 27.

H. Baxter was the first to publish this proposal⁷ and his 1992 book, *A Handbook of Old Chinese Phonology*, employs statistical methods to prove that six vowels account for the rhymes of the 詩經 *Shījīng* better than previous systems.

Since around the turn of the last millennium Baxter has worked with Laurent Sagart to refine and augment the 1992 system. The work under review, together with an accompanying online spreadsheet,⁸ provides a synthesis of their collaboration so far. The passing of 22 years saw three major modifications: 1. a panoply of proposals related to the reconstruction of cluster onsets, argued for on the basis both of morphological hypotheses⁹ and of phonetic correspondences with proto-Min and Chinese loans into Hmong-Mien, Vietic, and Tai-Kadai languages, 2. the reconstruction of uvulars following a suggestion of Pan Wuyun,¹⁰ 3. the introduction of final *-r following a suggestion of Sergei Starostin.¹¹ These three modifications are well motivated and convincing. In particular, the consideration of proto-Min and early loans into non-Sinitic languages is an unambiguous step forward; all future researchers would do well to follow this example. Nonetheless, a certain lack of explicitness in the argumentation and the selective presentation of evidence leaves the reader seldom able to confirm the proposed reconstruction *ab initio*.

1. The hypothesis that Old Chinese may have enjoyed cluster onsets goes back at least to von der Gabelentz.¹² Baxter & Sagart primarily bring to bear the previously untapped sources, Proto-Min and early Chinese loanwords into non-Sinitic languages, in the reconstruction of such cluster onsets. They present the specific theory, earlier elaborated in Sagart's *The Roots of Old Chinese* (1999), that cluster onsets existed in both a "tight" and "loose" form; in a choice of terminology destined to contribute to historical Chinese phonology's reputation as a dark art, they refer to the initial of such onsets oxymoronically as a "pre-initial." In his earlier book, Sagart¹³ was at pains to elaborate the phonotactic

⁷ Baxter, "Some proposals on Old Chinese phonology," 1–33. Independently of Baxter, Starostin, *Реконструкция*, arrived at a similar system. Zhengzhang, *The Phonological System of Old Chinese*, 33–42, and Schuessler, *Minimal Old Chinese and Later Han Chinese*, also adhere to the six vowel hypothesis. Although the hypothesis has opponents, to my knowledge there exists no detailed rebuttal.

⁸ The spreadsheet is available at <http://ocbaxtersagart.lsa.umich.edu/> (accessed 31 October 2015).

⁹ Space considerations preclude the discussion of morphology here.

¹⁰ Pan, "喉音考," 10–24.

¹¹ Starostin, *Реконструкция*, 399–407.

¹² Von der Gabelentz, *Chinesische Grammatik*, 99, §223.

¹³ Sagart, *Roots of Old Chinese*, 13–19.

theory of “loose” and “tight” pre-initials, but much of the evidence he put forward then, such as that Cantonese *kə-la:k-twi* “armpit” and Fuzhou *kɔ-lou?-a* “id.” suggest a tight preinitial *k- in the word 脇 *kak* < *k.lʰak “armpit,”¹⁴ no longer holds in the new reconstruction; 脇 *kak* is now *[C.q]ʰak. A consequence of such unmentioned discrepancies between the current work and the authors’ previous research is that one must still regularly consult their earlier works, but with the misgiving that some (potentially key) details no longer hold.¹⁵

In support of their reconstructions of Old Chinese pre-initials Baxter and Sagart cite evidence from various Vietic languages including Maleng Brô, Sách, Pong, Rục, and of course Vietnamese. Fricative initials in Vietnamese sometimes corresponds to a “pre-syllable” in other Vietic languages. For example, Vietnamese *vải* “cotton,” *vôi* “lime,” and *vả* “hit” have the Thavung cognates *kpaasʰ*, *kpuulʰ*, and *tpahʰ* respectively;¹⁶ thus, in early Chinese loans Baxter and Sagart take Vietnamese spirantization as evidence of an original Chinese pre-initial (p. 47). However, because Vietnamese initial fricatives also have other origins, the use of Vietnamese initial fricatives as *prima facie* evidence of lost pre-syllables is insufficient. For example, initial *v-* descends from *w- as well as *Cp- / *Cb-; these two origins are still distinct in Alexandre de Rhodes’ (1651) *Dictionarium Annamiticum Lusitanum et Latinum*.¹⁷ To preclude *w- as the source of *v-* in a particular word Baxter and Sagart should have cited the relevant form in de Rhodes’ dictionary rather than the modern spelling.

The scant attention that Baxter and Sagart give to Vietic languages other than Vietnamese leaves the reader to puzzle over the idiosyncrasies of the Vietic words they cite. Forms such as Pong /ktu:j/ “mallet” and Rục /kæci:ŋ 2/ “bed” show what one might call “tight” and “loose” pre-initials, but Baxter and Sagart make no attempt to account for this Vietic distinction.

2. Pan proposes uvulars largely on the basis of Sino-Tibetan parallels and early transcriptions of foreign words, but points out resulting improvements in the behaviour of phonetic series.¹⁸ On the basis of series that mix readings with Middle Chinese velars (見 *k-*, 群 *g-*, 匣 *h-*, 曉 *x-*) and readings with initial glottal stop (影 ‘-’) or initial 以 *y-*, Baxter and Sagart propose that the fricative (匣 *h-*,

¹⁴ Ibid., 14–15.

¹⁵ Baxter’s 1992 reconstructions of Old Chinese were also more predicative than the current version. For example, the previous system represented the failure of velars to palatalize before front vowels, e.g., 吉 *kjit* < *kjit (*Handbook*, 213); the new reconstruction fails to notate this development, e.g., 吉 *kjit* < *C.qi[t].

¹⁶ Ferlus, “Spirantisation des obstruantes,” 98.

¹⁷ Ibid., 89–90.

¹⁸ Pan, “喉音考,” 10–24.

曉 *x-*), glottal (影 ‘-’) and glide (以 *y-*) initials found in such series originate from uvular initials (pp. 43–46). They explain velar stops (見 *k-*, 群 *g-*) occurring in the relevant series as arising from complex onsets (pp. 168–70),¹⁹ e.g., 公 (𠂔 in early texts) *kuwng* < *C.qʰ-, 瓮 *‘uwngH* < *qʰ-, 容 (𠂔) *yowng* < *Gr-. Unfortunately, their proposals are not sufficiently explicit to allow the reader to predict their reconstruction of a given word. For example, they reconstruct many of the readings in the series built on 貴 with velar initials (e.g., 簣 *khweajH* < *kʰr- and 贍 *ngweajH* < *[ŋ]ʰr-), despite the presence of readings such as 漬 *hwojH* and 遺 *ywij* suggesting a uvular series.²⁰ Their proposed developments of uvulars occasionally appear phonetically odd. For example, they propose that *m.qʰ-, *m.G-, *N.qʰ-, *N.G- develop to 疑 *ng-*, whereas *m.q- and *N.q- merge with *G-. Nonetheless, the clearcut correspondence of Chinese uvulars to Tibetan velars and Burmese zero onset (e.g., Chi. 箴 *tsyim* < *t.qəm “needle,” Tib. ཁམ་ *khav*, Bur. འཕྱུག་ *ap*; Chi. 窞 *‘imH* < *q(r)[ə]m-s “subterranean room,” Tib. རྒྱལ་ *khyim* “home,” Bur. རྒྱལ་ *im*; Chi. 縊 *‘ejH* < *qʰik-s “strangle,” Tib. རྒྱལ་ *hkhuyig* “tie, fasten, suffocate,” Bur. རྒྱལ་ *ac* “squeeze, throttle,” etc.) versus velars in all three languages (Chi. 苦 *khux* < *kʰaʔ “bitter,” Tib. ཁ་ *kha*, Bur. ཁ་ *khāh*; Chi. 岡 *kang* < *kʰaŋ “hill,” Tib. སྒོ་ *sgaŋ*, Bur. སྒོ་ *khaŋ*, etc.) convincingly confirms the correctness of the Old Chinese uvular hypothesis.

3. Starostin motivated final *-r in Old Chinese, primarily on the basis of phonetic series that mix reading that have both final *-an* 元 and (*aj >) *-a* 歌 in Middle Chinese.²¹ Baxter and Sagart argue that the evidence of such series coincides with early rhyme contacts of *-n* and *-j and transcriptions of /r/ in foreign languages. Unfortunately, they fail to tell the whole story. On the one hand they suppose that change of *-r to *-n* occurred early; 𠂔, which they reconstruct *s-ŋrar, is used in two transcriptions they leave unmentioned, viz. 貴山 “Kushan” (59 BCE) and 烏弋山離 “Alexandria” (36 BCE). If this character originally had an *-r reading, this *-r must have changed to *-n* before 59 BCE to be suitable in these transcriptions. On the other hand, they accept Lokakṣema’s even later (178–89 CE) use of the character 𠂔 to transcribe Indic *-svar-* in the writing of Skt. *ābhāsvara* as 阿會

¹⁹ In an earlier paper, Sagart and Baxter, “Reconstructing Old Chinese uvulars,” 231, proposed that “loose” pre-initials conditioned the shift from uvulars to Middle Chinese velars, but because some of the relevant words do not have “softened” initials in proto-Mǐn they have revised their proposal to instead suggest “tight” pre-initials before uvulars are the source of the relevant cases of velars (p. 24).

²⁰ In another potential instance of this lack of explicitness, Baxter and Sagart include *t.qʰ- > 端 *t-* in the table that summarizes the correspondences involving *t- (p. 157), but I am unable to locate an explicit discussion of this change in their volume.

²¹ Starostin, *Реконструкция*, 399–401.

𠬞 (p. 258)²² as evidence for final *-r in this phonetic series, despite the fact that the same translator in the very same text uses 𠬞 for -vāṇ- in the transcription of *parinirvāṇa* as 般泥洹 (178–89 CE)²³ and 桓 for -vān- in the transcription of *śakro devānām indra* as 釋提桓因.²⁴ If *-r had become -n by 59 BCE, then it was also -n in 178 CE. If Lokakṣema maintained -r in his speech for some words he ought to have maintained it for all.²⁵ Thus, although the overall thrust of Baxter and Sagart's discussion strongly suggests *-r as part of the Old Chinese syllable canon, their cherry-picking of the evidence undermines confidence in any one of their reconstructions.

This review airs some criticisms, but it is easy to behold the mote in a brother's eye. I have learned an enormous amount from this book, which irrevocably improves the study of Chinese phonology. Future research will meticulously fill in the lacking details, but until then the new ideas may be slow to catch on.

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²² Coblin, *Handbook of Eastern Han Sound Glosses*, 244 #68.

²³ *Ibid.*, 243 #51.

²⁴ *Ibid.*, 243 #52.

²⁵ Other explanations include that Lokakṣema retained *-r only in restricted phonetic environments or that the presence of dialect loans in his dialect gave rise to a patterning of -n versus *-j readings distinct from Middle Chinese. The matter requires exploration before his transcriptions are useful to support *-r in Old Chinese.

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